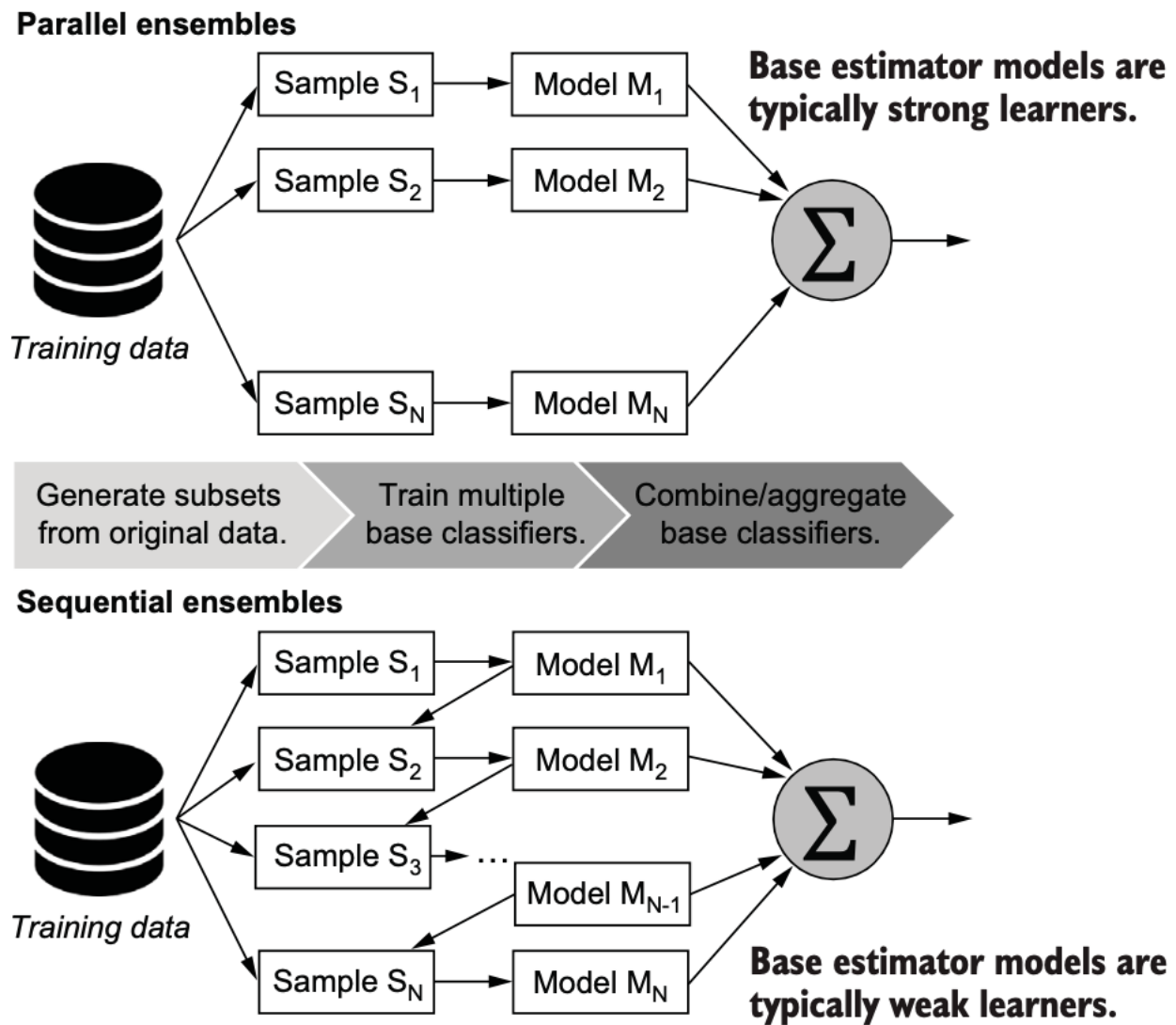


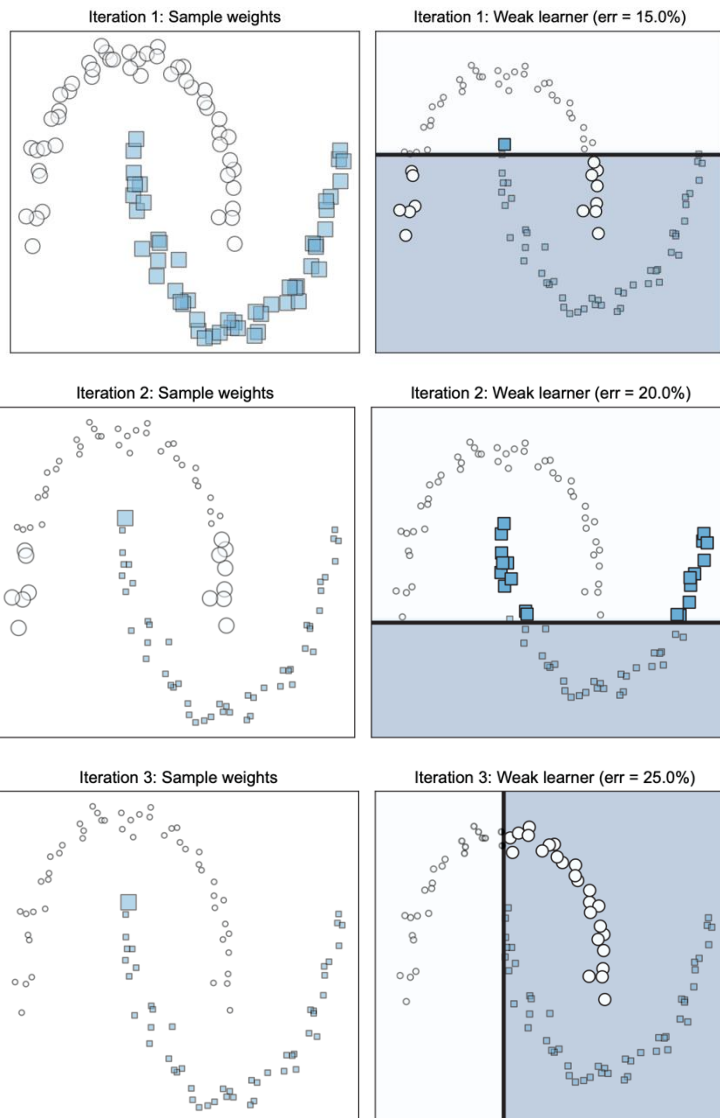
Boosting

2026.04.09

Parallel vs sequential ensembles

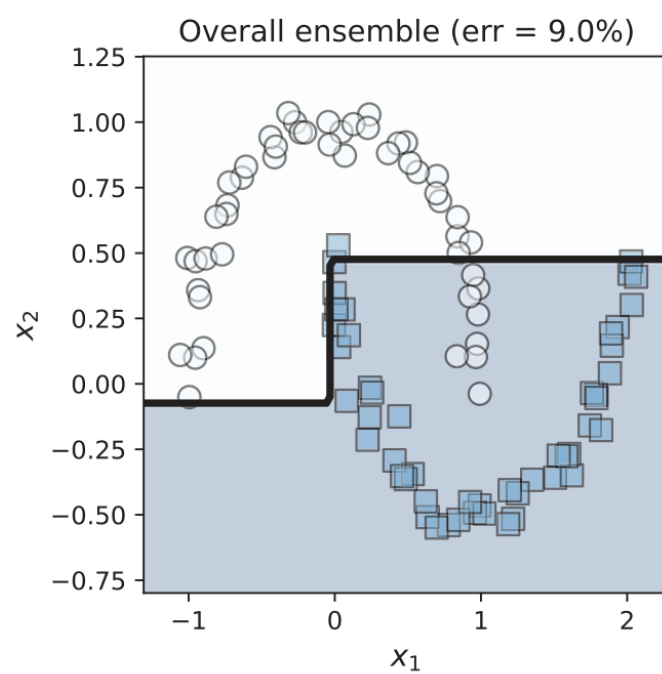


AdaBoost



$$\text{weighted error} = \sum_i w_i \cdot \mathbf{1}(y_i \neq \hat{y}_i)$$

AdaBoost



AdaBoost

When to use?

XGBoost

- Tree based
- Loss function (critical) and gradient descent
- Learning rate
- Can bootstrap (is not default, subsample)
- Can handle some missing values
- Faster as individual tree features evaluated in parallel
- Smart pruning (gamma)
- Early stopping

XGBoost

When to use?